

Short Communication: Effect of Subacute Ruminal Acidosis on In Situ Fiber Digestion in Lactating Dairy Cows

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ABSTRACT

Subacute ruminal acidosis (SARA) was induced by replacing 25% of the total mixed ration intake [dry matter (DM) basis] with pellets consisting of 50% wheat and 50% barley. This reduced dietary forage content (DM basis) from 39.7 to 29.8% and increased the dietary concentrate content from 60.3 to 70.2%. Induction of SARA reduced the 24- and 48-h in situ neutral detergent fiber (NDF) degradabilities of grass hay numerically from 31.5% to 24.6% ($P = 0.29$) and from 51.3% to 36.9% ($P < 0.05$), respectively. The 24- and 48-h in situ NDF degradabilities of legume hay were reduced from 35.3 to 26.3% ($P < 0.05$) and from 49.0 to 35.8% ($P < 0.05$), respectively. The 24- and 48-h in situ NDF degradabilities of corn silage were reduced from 44.0 to 37.2% ($P < 0.05$) and from 56.1 to 44.8% ($P < 0.05$), respectively. This study suggests that induction of SARA by excess feeding of wheat/barley pellets reduces the rumen digestion of NDF from grass hay, legume hay, and corn silage.

(Key words: subacute ruminal acidosis, fiber digestibility, dairy cows)

Abbreviation key: NFC = nonfibrous carbohydrate; SARA = subacute ruminal acidosis.

INTRODUCTION

Subacute ruminal acidosis (SARA) is a common and economically important metabolic disorder in dairy cattle (Nocek, 1997; Garret et al., 1999). Cows experiencing this metabolic disease have daily episodes of low rumen pH that impair microbial function, especially the function of fibrolytic microflora (Shi and Weimer, 1992). Due to the difficulty in diagnosing SARA, limited information on the incidence of this disorder exists (Garrett

et al., 1999). However, Oetzel et al. (1999) found an incidence of SARA of 20.1% in cows between 1 and 120 DIM on commercial dairy farms in Wisconsin. SARA also causes decreased appetite, diarrhea, and laminitis (Nocek, 1997). This metabolic disorder can be the result of excess feeding of nonfibrous carbohydrates (NFC), a rapid increase in the dietary content of NFC, or insufficient rumen buffering (NRC, 2001). After calving, a drop in rumen pH resulting from an accumulation of VFA can be expected, due to the higher concentrate and a lower effective fiber content of the postcalving diet, resulting in a higher rate of VFA production and the reduction of the absorptive capacity of the rumen mucosa for VFA during the dry period (Dirksen et al., 1984). SARA has been defined by rumen pH values between 5.2 and 5.6, whereas values below 5.2 signify acute acidosis (Cooper and Klopfenstein, 1996). Garret et al. (1999) suggested a cutoff point of rumen pH 5.5 for diagnosis of SARA by rumenocentesis. It has been shown that in vitro fiber digestibility is reduced when pH drops below pH 6.2 (Grant and Mertens, 1992; Grant, 1994; Calsamiglia et al., 1999). Hence, the rumen pH below which SARA occurs is subject to controversy. Varel and Kreikemeier (1995) concluded that in vitro digestion methods do not adequately reflect the effect of low rumen pH on NDF digestion in the rumen. Hence, these studies might not be fully representative for the effect of SARA on fiber digestion.

The objective of this study was to determine the effect of inducing SARA by excess grain feeding on the in situ digestion of grass hay, legume hay, and corn silage.

MATERIALS AND METHODS

Four multiparous ruminally fistulated Holstein dairy cows were housed in individual tie stalls at the Elora Dairy Research Centre throughout the experiment, in accordance with the guidelines of the Canadian Council of Animal Care. The experiment followed a crossover design with two periods of 2 wk. During the second week of each period, SARA was induced in two cows

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for five consecutive days. Two other cows received TMR for ad libitum intake mixed with 100 g of sodium bicarbonate each day (control). In the subsequent period, treatments were switched. SARA was induced by replacing 25% of the ad libitum TMR intake (DM basis) with an equal amount of DM from grain pellets consisting of 50% ground wheat and 50% ground barley, feeding 2 kg of TMR at 7 a.m. and only providing access to TMR between 11:00 and 11:30 a.m., between 3:00 and 3:30 p.m. and after 5 p.m. TMR consisted of (DM basis) 18.1% corn silage, 28.4% high moisture corn, 22.3% mixed haylage, 6.6% mixed hay and 24.7% lactating cow supplement. The DM content of TMR was 50.4%. TMR contained (DM basis) 39.7% forage, 60.3% concentrate, 17.2% CP, and 29.6% NDF. The lactating cow supplement contained (as-fed basis) 3.9% blood meal, 8.6% corn gluten meal, 20.6% soybean meal, 2.65% dicalcium phosphate, 2.94% calcium carbonate, 1.03% magnesium oxide, 1.77% salt, 0.88% Dynamate (22% S, 18% K, 11% Mg; IMC Feed Ingredients, Lake Forest, IL), 0.44% Dairyman's Edge (LBJ Pakke Inc., Waddington, NY), 0.27% FFM Micro premix (Dicalcium phosphate, calcium carbonate, magnesium oxide, trace minerals, vitamin A, vitamin D, vitamin E; Floradale Feedmill, Floradale, ON), 8.8% wheat shorts, 2.2% fish meal, 22.1% soybean hulls, 3.7% molasses, 0.2% Alimet (DL-Methionine Hydroxy Analogue; Novus, St. Louis MO), 7.4% wheat grain, and 11.0% roasted soybean. All cows also received 1 kg d⁻¹ of a pellet containing (as-fed basis) 30% dehydrated alfalfa, 30% ground soy hulls, and 40% wheat shorts. Promote N.E.T (Agribrands International,) was added to the pellets, providing xylanase and cellulase. Enzyme activity, measured as micromoles of reducing sugars from cellulose per minute (pH 5.0, temperature 40°C) was 1200 ACU g⁻¹ for cellulase. Through the enzyme pellets cows received 4 g of active enzyme per day. During control periods, the DM of the diet contained 39.7% forage and 60.3% concentrate. During SARA periods, the DM of the diet contained 29.8% forage and 70.2% concentrate. Cows had unlimited access to fresh water.

Feed intake and milk yield were monitored throughout the experiment. Morning and afternoon samples of milkings on Mondays and Thursdays were pooled and were sent immediately for analysis of fat, protein, and lactose by near infrared analysis using the Foss System 400 (Foss Electric, Hillerød, Denmark) to the Laboratory Services Division of the University of Guelph (Guelph, ON, Canada).

Rumen pH was measured continuously throughout the experiment using indwelling pH probes, as described by Cumby et al. (2000). Rumen pH was measured every second, but only averages over 60 s were stored and used for subsequent analysis. Rumen pH

data were summarized by calculating average pH, time below pH 6 and time below pH 5.6, area (time × pH) below pH 6, and area (time × pH) below pH 5.6 for each 24-h period. Averages for each period were analyzed using the SAS general linear models procedure (SAS, 1990). The model included cow, period, and treatment as fixed effects. A probability of less than 0.05 was considered significant.

In situ rumen digestibility measurements of three test feeds, i.e., alfalfa hay, grass hay, and corn silage, were determined as described by Ørskov et al. (1980). Air-dried test feeds were ground through a 1-mm screen using a Wiley mill. Nitrogen-free polyester bags with a pore size of 50 microns (ANKOM, Fairport, NY) containing 3 g of each ground-dried test forage were incubated in the rumen for 3, 6, 12, 24, 48, and 72 h starting at 9 a.m. of the second day that SARA was induced for SARA cows and control cows. After each incubation time, the duplicate bags were removed and hand washed in lukewarm water for 5 min. At the beginning of the trial, four bags containing 3 g of test feed were soaked in warm water for 1 h, and then washed to determine washing losses. The residues in the bags were then dried for 48 h at 60°C and the DM loss was determined. The NDF was determined as described by Goering and Van Soest (1970). The legume hay had a composition (DM basis) of 19.9% CP, 46.4% NDF, and 37.0% ADF. The grass hay had a composition (DM basis) of 12.3% CP, 64.1% NDF, and 38.9% ADF. The corn silage had a composition (DM basis) of 12.5% CP, 36.0% NDF, and 22.0% ADF. The rumen DM and NDF digestibilities were fitted to an exponential equation using the SAS nonlinear regression procedure (SAS, 1990). Calculated parameters included potential total digestibility (A + B) and digestibility rate (c), digestibility at 24 h (24 h), and digestibility at 48 h (48 h).

RESULTS AND DISCUSSION

Induction of SARA significantly reduced average daily rumen pH from 6.36 ± 0.17 (mean $\pm$ SE) to 5.72 ± 0.17 ($P < 0.05$), significantly increased daily time below pH 6 from 154.8 ± 174.1 min d⁻¹ to 937.4 ± 174.1 min d⁻¹ and daily time below pH 5.6 from 15.4 ± 188.9 min d⁻¹ to 594.4 ± 188.9 min d⁻¹, and numerically increased area below pH 6 from 29.3 ± 159.5 min*pH d⁻¹ to 518.4 ± 159.5 min*pH d⁻¹ ($P = 0.07$) and area below pH 5.6 from 5.3 ± 88.8 min*pH d⁻¹ to 228.0 ± 88.8 min*pH d⁻¹ ($P = 0.12$).

The ad libitum intake of TMR was 17.8 kg DM d⁻¹ during control periods, and the TMR intake was 11.4 kg DM d⁻¹ during SARA periods. This shows that during the periods that SARA was induced, the cows did not consume the provided 75% of the ad libitum TMR intake

Table 1. Effect of subacute ruminal acidosis (SARA) on in situ ruminal NDF digestion of test forages.

	SARA	Control	SE	P
Corn silage ¹				
A + B (%)	54.2	66.3	3.49	NS ²
c (% h ⁻¹)	2.5	1.8	0.7	NS
24 h (%)	37.2	44.0	1.1	<0.05
48 h (%)	44.8	56.1	1.3	<0.05
Grass hay ¹				
A + B (%)	50.2	66.3	3.3	<0.05
c (% h ⁻¹)	2.9	2.0	0.1	<0.05
24 h (%)	24.6	31.5	4.0	NS
48 h (%)	36.9	51.3	4.2	<0.05
Legume hay ¹				
A + B (%)	43.4	54.3	2.3	NS
c (% h ⁻¹)	4.0	3.8	0.1	NS
24 h (%)	26.3	35.3	0.9	<0.05
48 h (%)	35.8	49.0	2.3	<0.05

¹A + B = Potential total digestibility; c = digestibility rate; 24 h = digestibility at 24 h; 48 h = digestibility at 48 h.

²Nonsignificant ($P \geq 0.05$).

during control periods. Average daily milk yield during control periods (31.1 kg d^{-1}) was not significantly different from that during SARA periods (31.5 kg d^{-1}). Milk fat content during control periods (3.43%) was higher than that during SARA periods (3.03%), but this difference was not statistically significant. Milk protein content and milk lactose content were 3.11 and 4.76%, respectively, during control periods and were 3.03 and 4.77%, respectively, during SARA periods, and were not affected by the induction of SARA. Hence, apart from the numeric reduction in milk fat content, SARA did not affect animal productivity. This could have been due to the small number of cows included in the experiment or to the short duration of SARA periods, i.e., only five consecutive days. Hence, any challenges to animal health caused by SARA were only present for a short period, after which animals were given the opportunity to recover.

Induction of SARA reduced the 24- and 48-h NDF digestibilities of all three test feeds (Table 1). Average rumen pH during SARA was slightly higher than the constant pH of 5.7 in Calsamiglia et al. (1999), whereas the average rumen pH during the control periods was close to the constant pH of 6.4 in Calsamiglia et al. (1999). In Calsamiglia et al. (1999), in vitro NDF digestion decreased from 53.8% at the constant pH of 6.4 to 37.4% at the constant pH of 5.7. This decrease is similar to the reduction in 48-h digestion of grass hay and legume hay but lower than the reduction in NDF digestibility of corn silage due to induction of SARA observed in the current study. However, a problem with the comparison of the results from the current study with those obtained by Calsamiglia et al. (1999) is that those authors determined NDF digestion of a diet containing

both alfalfa hay and concentrate, while the current study only determined NDF digestibility of forages.

Average rumen pH during SARA periods was close to the lowest pH included in the in vitro experiments of Grant and Mertens (1992) and Grant (1994). However, in the current experiment, rumen pH showed much more post-feeding variation than in the earlier in vitro studies. The reduction in 48-h NDF digestibility of legume hay observed in the current study was smaller than the reductions in NDF digestion of alfalfa hay by reducing incubation from pH 6.2 to pH 5.5 with and without addition of cornstarch (Grant, 1994). However, the reduction in NDF digestion of grass hay in the current study was greater than that observed by Grant (1994). These differences could be due to the larger reduction in rumen pH in the earlier study and the effect of forage source on the depression in NDF digestion due to pH reduction. The latter could be explained by differences in the composition of the NDF fraction between forages. The current results were moderately higher than the reduction in NDF digestion of alfalfa hay, bromegrass hay, and corn silage due to reduction of in vitro pH from 6.2 to 5.8 (Grant and Mertens, 1992). This could be due to the extent of digestion generally being greater for in situ than for in vitro methods of determining NDF digestion (Varel and Kreikemeier, 1995), or due to a constant pH of 5.8 being less detrimental to fiber digestion than the diurnal fluctuations in rumen pH around 5.72 during SARA in the current experiment.

Also, in the in vitro studies (Grant and Mertens, 1992; Grant 1994; Calsamiglia et al., 1999) rumen pH was set at a constant level or varied using buffers, inorganic acids, and sodium hydroxide. This creates different fer-

mentation conditions than in the current *in vivo* experiment, where diurnal variation in the rumen was due to changes in the rumen concentrations of VFA and buffers from the saliva. Hence, in the current study, a more natural simulation of SARA was achieved.

The addition of the enzyme mix did not prevent the reduction in fiber digestion due to the induction of SARA. As all cows received the enzyme mix during all experimental periods, no conclusion of the effect on the enzyme mix on fiber digestion can be drawn. However, the observed depression in fiber digestion due to a rumen pH that is low, but still within the physiological range observed in high-yielding dairy cows on high-concentrate diets, indicates that fiber digestion in these cows needs to be increased.

Induction of SARA through excess feeding of grain pellets for five consecutive days resulted in a significant reduction in the *in situ* NDF digestion of corn silage, grass hay, and legume hay and a numeric reduction of milk fat content. This reduction in *in situ* NDF digestion was higher than what could have been predicted from earlier *in vitro* studies. Inducing SARA did not affect milk yield and milk protein percentage. As the presence and source of starch and the duration of SARA will affect fiber fermentation and measures of animal health, the effects of other *in vivo* models on inducing SARA during prolonged periods, e.g., through reducing dietary particle size, warrant investigation. Additionally, the effects of SARA on fiber digestion of entire diets need to be confirmed.

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